testlio

AI, Accessibility, and the Art of Deception!

Where AI shines, where it fails, and why expertise matters.

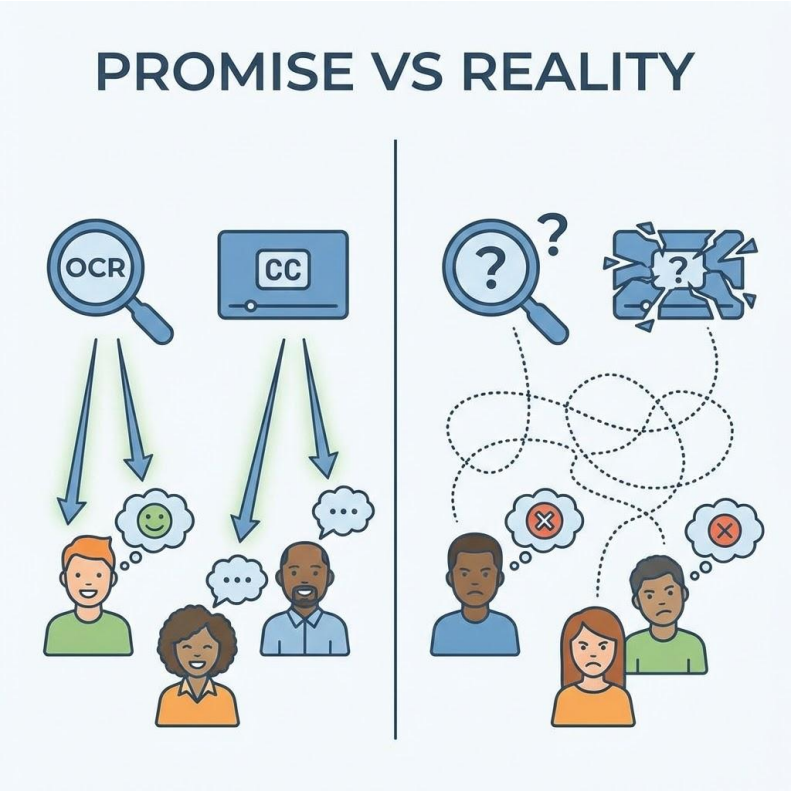
Testlio Confidential 2026

Foundations of AI and Accessibility

AI in Accessibility: Promise versus Reality

AI can be incredibly powerful for accessibility, but only when used with knowledge, oversight, and skepticism.

- **AI's Accessibility Benefits**
- **Limitations of AI in Accessibility**
- **Human Expertise Importance**
- **Balancing Promise and Reality**



Where AI Helps Assistive Technology Users

AI technologies provide significant value when applied directly to enhance the experience of users who rely on assistive technologies such as screen readers, magnifiers, switch devices, and voice interfaces.

- **AI-Powered Text Recognition**
- **Automated Alt-Text Suggestions**
- **AI-Driven Captioning Systems**
- **Enhanced Navigation and Interaction**

AI for Testing – Focus Areas

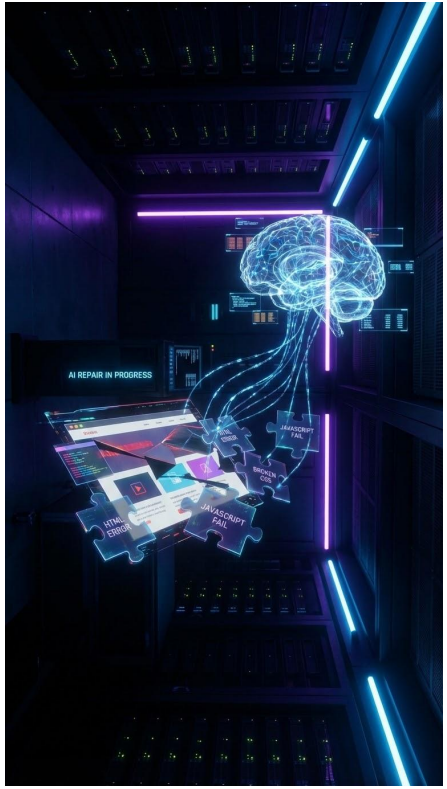
AI can rapidly analyse test results, logs, and trends, reducing time spent on reporting and insight generation.

- **Faster analysis and reporting**
- **More reliable automated test suites**
- **Improved context for testers**
- **Accelerated test case design (with limits)**

AI Compensating for Poorly Coded Websites

In environments where digital content is poorly constructed or lacks basic accessibility support, AI can play a valuable compensatory role by providing users with additional layers of interpretation and assistance.

- **AI Enhancing Accessibility**
- **Content Interpretation and OCR**
- **Improved Navigation and Layout**
- **Limits and Roles of AI**

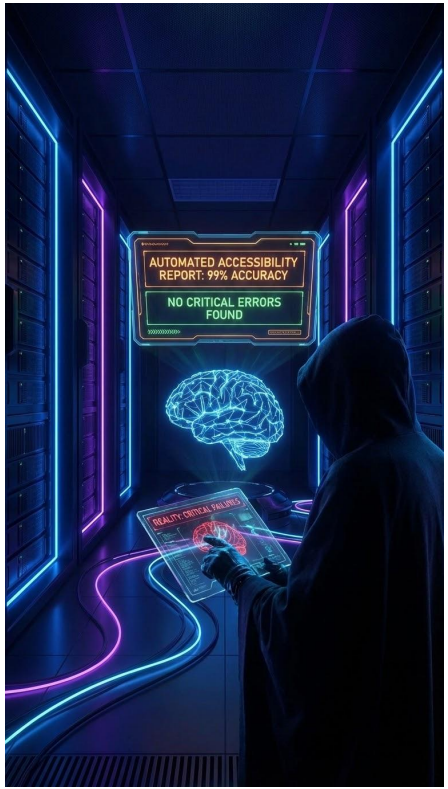


Limitations and Risks of AI in Accessibility

The Art of Deception in Automated Testing

AI-based automated accessibility testing tools often create an illusion of thoroughness and accuracy, leading teams to believe they have achieved a high degree of compliance when they have not.

- **Illusion of Thoroughness**
- **Lack of Contextual Understanding**
- **False Positives and Negatives**
- **Risks of Over Reliance**



Case Study: Automation Tool Failure

A real-world case study illustrates the severe limitations of AI when used for accessibility testing. In this scenario, an AI-driven automated tool scanned a website and reported 300 issues

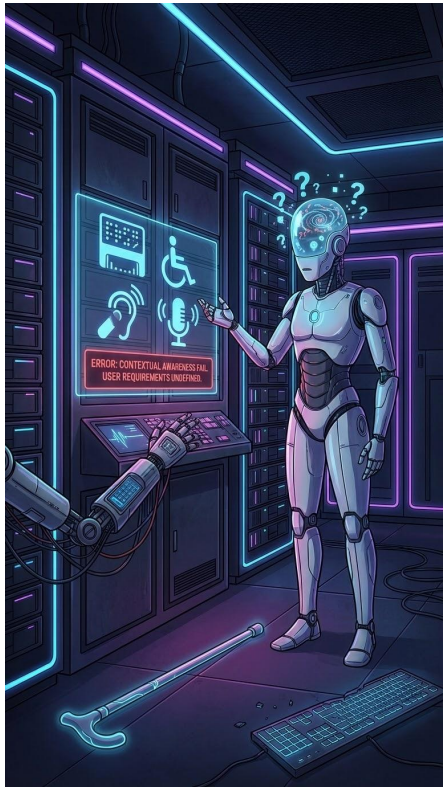
- **High False Positive Rate**
- **Low Accuracy in Findings**
- **Need for Expert Oversight**
- **Limitations of AI Tools**

Stage	Count	Accuracy
Issues reported by AI	**300**	-
Non-accessibility issues	**150**	❌
Remaining potential issues	**150**	?
Valid accessibility issues	**12**	✔️ (~4%)
Issues reported accurately	**6**	✔️ (~2%)

Why AI Struggles With Accessibility Context

AI struggles with accessibility because the discipline requires a level of contextual and experiential understanding that exceeds the capabilities of current automated systems.

- **Contextual Understanding Limitations**
- **Dynamic Interaction Challenges**
- **Neurodivergent Accessibility Factors**
- **Limitations of Automated Evaluation**

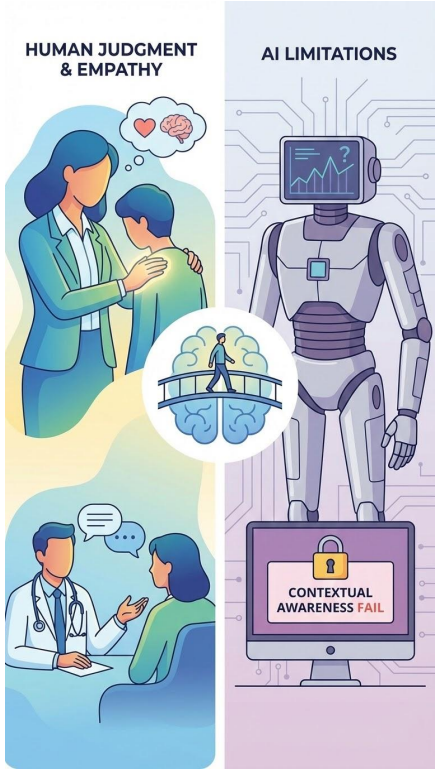


Expert Oversight and Manual Testing

Why Human Expertise Remains Essential

Human expertise is central to successful accessibility evaluation because it brings judgment, empathy, and contextual awareness that AI fundamentally lacks.

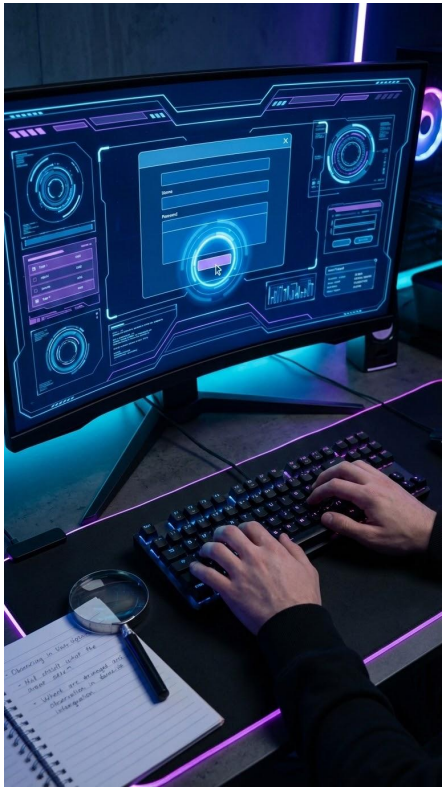
- **Human Judgment and Empathy**
- **Understanding Usability Barriers**
- **Interpreting Assistive Technology Nuances**
- **Ethical and Holistic Evaluation**



Why Manual Testing Will Always Be Required

Manual testing is an irreplaceable component of accessibility evaluation because it assesses aspects of user experience that AI cannot capture.

- **Human Insight Beyond AI**
- **Keyboard and Focus Testing**
- **Neurodivergent Accessibility Assessment**
- **Validation of AI Suggestions**

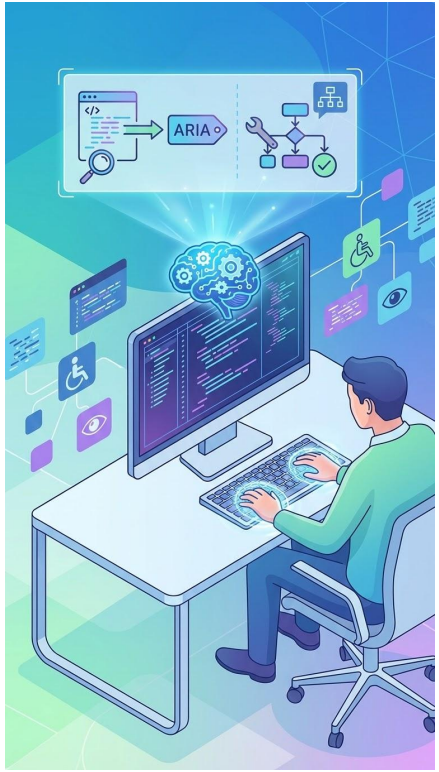


Best Practices for Using AI Responsibly

Effective and Responsible Use of AI

AI can significantly enhance accessibility workflows when used responsibly and with appropriate limitations.

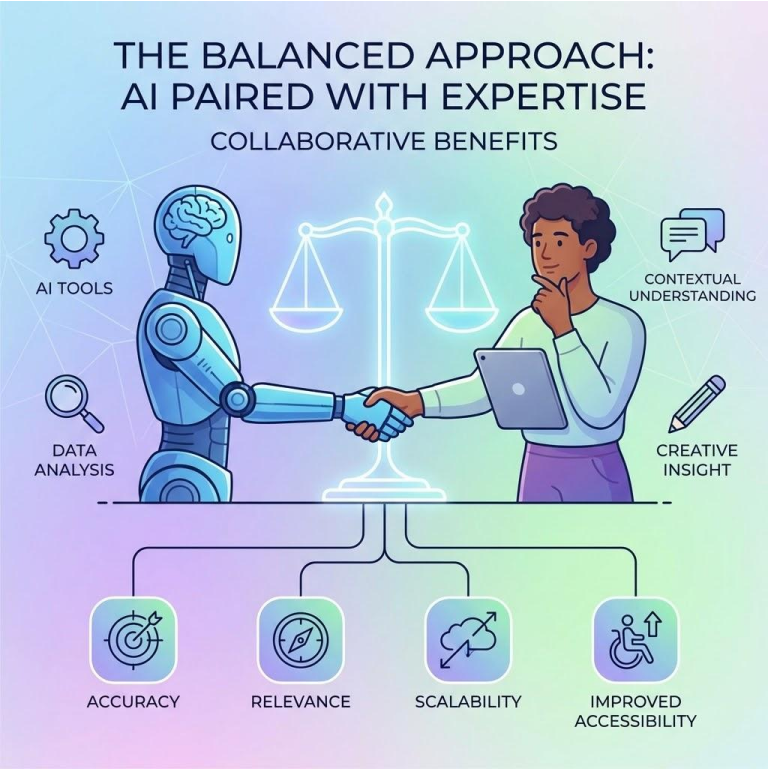
- **AI Enhances Accessibility Workflows**
- **Developers Benefit from AI Tools**
- **Responsible AI Use Principles**
- **Integration into Broader Strategy**



The Balanced Approach: AI Paired With Expertise

The most effective accessibility strategy combines AI capabilities with the nuanced understanding of human experts.

- **AI Strengths**
- **Human Expertise**
- **Collaborative Benefits**



Conclusion and Forward Outlook

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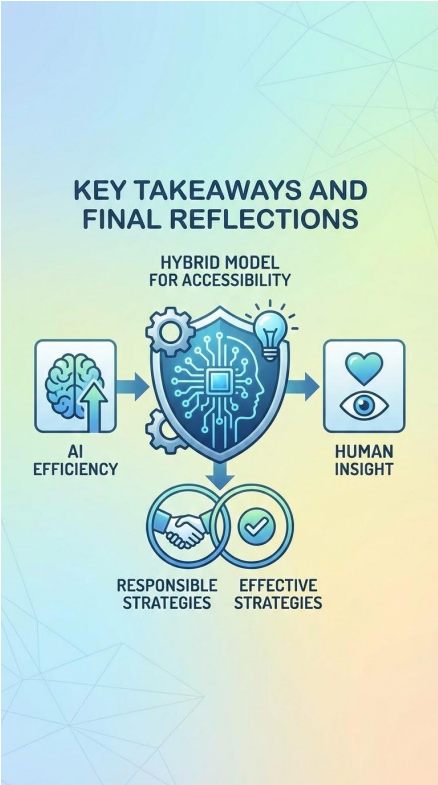
When AI and accessibility converge, we build workplaces that are more productive, innovative, and human-centered.”

Neil Milliken

Key Takeaways and Final Reflections

AI offers powerful capabilities that can support accessibility, enhance user experiences, and assist developers in identifying potential issues. However, its limitations are significant, particularly when it is used for accessibility testing or compliance verification.

- **AI's Accessibility Support**
- **Limitations of AI in Accessibility**
- **Human Expertise is Essential**
- **Hybrid Model for Accessibility**



Thank you!

Crowdsourced testing, powered by AI.